

Invited: 2026 Extensions of IEEE CEDA DATC's Robust Design Flow and Other Activities

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Abstract

Over the past year, IEEE CEDA DATC has continued to improve the DATC Robust Design Flow (RDF) while expanding research infrastructure, benchmarking foundations, and AI-enabled EDA methodologies for the global research community. This paper highlights four directions: (1) extension of the first public commercial EDA benchmarking effort from netlist-to-GDS to RTL-to-GDS; (2) new and expanded research infrastructure and enablements; (3) expansion of DATC activities into new areas; and (4) advances in AI-enabled EDA. The paper also reviews community and ecosystem activities and provides an updated RDF roadmap.

CCS Concepts

• **Hardware** → **Electronic design automation.**

Keywords

electronic design automation, research infrastructure, benchmarking, AI-enabled EDA

ACM Reference Format:

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1 Introduction

The Design Automation Technical Committee (DATC) within the IEEE Council on EDA (CEDA) [42] seeks to strengthen the foundations on which design automation research is developed, evaluated, and shared. A central goal is to lower barriers to high-quality and reproducible EDA research via accessible design environments, meaningful benchmarks and baselines, common metrics, technology enablements, and mechanisms for sharing and reproducing new methods. One of DATC's longest-running efforts toward this goal is the Robust Design Flow (RDF). RDF was introduced in 2016 [9] to integrate and preserve academic EDA tools and enable their use in

YEAR	RDF Updates	Research Infrastructure & Enablement	Benchmarking & Baselines	AI-Enabled EDA
2016–20	RDF-2016 OpenDesign Flow Database grows into a full RTL→GDS flow; RDF-2020 adds integrated OpenROAD			
2021	Chisel/FIRRTL; RTL obfuscation; DFT	METRICS2.1; AutoTuner	RosettaStone; calibration	
2022	Timing-sensible RosettaStone	METRICS2.1 in ORFS; cloud DoE	Macro placement benchmarks	ML-driven macro placement
2023	TritonPart; PROBE3.0; 3D PD	Proxy design enablement	Partitioning; macro placement	Artificial netlists; IR-drop ML; CircuitOps data
2024	DG-RePIAce; MBFF; OpenSTA calibration	BEOL, multi-PVT tuning	Calibration leaderboard	ML data formats; LLMs for PD; artifact evaluation
2025	Continuous integration	OpenROAD-Research, ORFS-Research established	An EDA benchmarking first: Aprisa, netlist→GDS	ML EDA Commons; agentic EDA introduction
2026	Difetto; PD watermarking	GT2N platform; 3D OpenDB; research repository updates	Aprisa benchmarking extension to RTL→GDS	Agentic flow opt; autonomous code evolution; EDA-Schema 2.1

Fig. 1: Trajectory of IEEE CEDA DATC activities.

realistic design contexts [40]. RDF provides a common environment in which research tools can be exercised with other parts of the design flow. RDF continues to evolve as new tools and capabilities are contributed. Over time, DATC activities have also expanded beyond the components included directly in RDF. They now span research enablement and infrastructure, benchmarking and baselines, and AI-enabled EDA.

Ten years after the introduction of RDF, Figure 1 summarizes this evolution, with emphasis on developments since 2021. METRICS2.1 and AutoTuner introduced standardized metrics reporting and automated design-space exploration [12]. RosettaStone and calibration efforts provided common designs and reference results [13]. From 2022 through 2024, these efforts expanded through OpenROAD-flowscripts (ORFS) integration, improved calibration methodologies, and a calibration leaderboard [41]. In parallel, DATC increasingly supported ML-enabled EDA through datasets, infrastructure, and new methodologies [14, 15]. A major benchmarking milestone came in 2025 [16]. DATC reported the first public benchmarking of a commercial place-and-route tool with vendor permission, using Siemens EDA Aprisa on open testcases. OpenROAD-Research [38] and ORFS-Research [39] were also established as research-oriented “contrib” repositories. The 2026 activities build directly on these foundations. We first highlight this year's RDF updates, then describe broader DATC activities in the remainder of the paper.



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1.1 RDF 2026 Updates

The DATC RDF continues to incorporate targeted additions and tool updates. This year's RDF updates include the following.

Design-For-Test (DFT). Automated, efficient DFT capabilities are necessary to identify faults and validate a chip after manufacturing. Difetto [2] defines a DFT extension to the LibreLane [3] implementation flow. The augmented flow starts at the synthesis stage and ends with additional signoff. During synthesis with Yosys, sequential logic elements are replaced with scan equivalents and a boundary-scan chain is constructed for I/O. OpenROAD then stitches scan chains using placement-aware 2-opt to reduce chain wirelength. Combinational logic functions are extracted into individual modules using a custom Yosys extension. These are fed into the Quaigh [4] logic analysis tool to perform automatic test pattern generation (ATPG). This process inserts faults into the netlist and uses SAT to build test vectors that activate these faults, compressing vectors opportunistically. Difetto validates the integrity of the scan chain after routing in OpenROAD. It finally sanity checks generated test vectors against the resulting chains using cocoth [7].

Standard-cell library characterization. Time-consuming post-layout cell library characterization and Liberty generation are practical limiters to deployment of new enablements. Libretto [5] is an open-source cell library characterization tool that produces Liberty power and timing models for use in P&R flows. Libretto is controlled using a custom scripting language. Key library information, including name, units and characterization conditions, is defined by the user, followed by individual cells. Libretto consumes netlist and SPICE models for each cell and produces SPICE decks that test all input transitions. These decks are fed to a simulator such as ngspice [6], with simulation outputs used to build NLDL tables. Currently, multi-output cells are not supported.

Physical design watermarking. Physical design (PD) results capture significant implementation effort and therefore constitutes valuable design intellectual property (IP). *PDMarks* [8] develops a Kerckhoffs-compliant watermarking framework for protecting PD IP against unauthorized reuse. The framework distributes ownership evidence across placement, clock tree synthesis (CTS), and routing. At the placement stage, watermarks are encoded through the relative ordering of key-selected nearby standard cells; at the CTS stage, through the sequential-fanout parity of selected leaf clock buffers; and at the routing stage, through statistically verifiable biases in the routing behavior of key-selected signal nets. All watermark instances, object selections, and target values are derived from a single secret key using domain-separated HMAC-SHA256 evaluations. Thus, the security of the framework depends only on the secrecy of the key rather than the secrecy of the watermarking algorithms. *PDMarks* is evaluated on Nangate45 (NG45) [20] and ASAP7 [19, 25] designs. Across the evaluated testcases, the all-stage watermark achieves an approximate joint coincidence probability below 10^{-32} while introducing only small power, performance and area (PPA) changes. A pull request implementing *PDMarks* in OpenROAD has been submitted [73].

1.2 Organization of Paper

The remainder of this paper is organized around the major areas of DATC activity in 2026: benchmarking and baselines, research infrastructure and enablement, and AI-enabled EDA. Our DATC benchmarking effort expands substantially by extending the Aprisa study from netlist-to-GDS to RTL-to-GDS, adding new testcases, and incorporating updated tool capabilities. These benchmarking

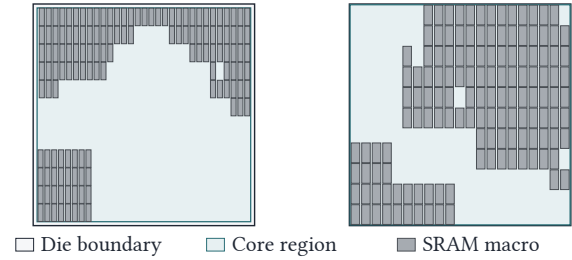


Fig. 2: *ariane* floorplans in ASAP7 (left) and NG45 (right). Each uses 136 SRAM macros and is input to both Aprisa and ORFS.

updates are described in Section 2. We also introduce new research enablements, including the 2nm GT2N platform and 3D IC design support through 3D OpenDB. These are discussed in Section 3. Section 4 highlights recent advances in AI-enabled EDA, including agentic optimization of complete ORFS flows and autonomous modification of EDA source code. Section 5 summarizes community and ecosystem activities, including workforce development efforts, contests, and hackathons, and Section 6 provides an updated roadmap for future DATC activities, including OpenROAD-Research [38] and ORFS-Research [39].

2 Public Benchmarking of Commercial SP&R

Public benchmarking makes leading-edge tool performance visible through reproducible comparisons. RDF-2025 [16] [43] performed the first public benchmarking (with vendor permission) of a commercial EDA P&R tool – Siemens EDA's Aprisa v2025.2 – using four open-source testcases in ASAP7 and NG45. In RDF-2026, we extend this benchmarking effort in three ways: (i) to Synthesis, Place and Route (SP&R) from P&R, (ii) from four to nine designs, totaling 18 design-PDK combinations, and (iii) to Aprisa v2026.1 [22] with the Crescent SP&R flow [23].

2.1 Common Inputs

Siemens EDA engineers have defined the common inputs for the benchmark. Aprisa v2026.1.1 and ORFS [26] are each provided with the same (i) RTL source; (ii) floorplan including core layout region and I/O placement; (iii) power plan; and (iv) Synopsys Design Constraints (SDC), including the target clock period (TCP). For designs with macros, the same macro placement is given to both RTL-to-GDS flows. Because synthesis is performed independently in each flow, the resulting cell area and placement utilization can differ.

The TCP for each design (see Table 1) is selected to expose setup-timing differences between Aprisa and ORFS. We first implement each design and PDK combination with Aprisa using an initial TCP value. If Aprisa reports zero setup (reg-to-reg) worst slack at this initial TCP, we tighten the TCP.

To support the benchmark SRAM configurations in both ASAP7 and NG45, we use a modified version of the original BSG Black-box SRAM Generator [27] that adds multi-port support [28]. We evaluate nine testcases with the ASAP7 and NG45 PDKs. Table 1 summarizes the benchmarks, macro counts, and evaluation configurations. Figure 2 shows the 136-macro *ariane* floorplan in each PDK.

2.2 Evaluation Methodology

Aprisa and ORFS independently perform SP&R using the common inputs specified in Section 2.1. Each flow produces (i) a routed DEF and (ii) a post-route gate-level netlist. These outputs, along with the input SDC, are provided to a third-party evaluation flow comprised of unnamed commercial P&R, signoff parasitic extraction (PEX), and static timing analysis (STA) tools. The P&R tool reports standard-cell count and routed wirelength for both Aprisa- and ORFS-generated layouts. The PEX and STA tools report power, worst negative slack (WNS), total negative slack (TNS), and the number of failing end-points (FEP). Design rule violation (DRV) and short counts cover routing violations only, and exclude DRVs induced by a limitation of the SRAM generators used [28].

Post-route parasitic extraction includes coupling capacitances, and the STA tool is invoked in signal integrity mode. For both ASAP7 and NG45, all input port drive resistances are set to 0.1 Ω ; output port capacitances are defined by post-route parasitics. Setup and hold WNS, TNS, and FEP are computed over register-to-register paths.

Reported total power is the sum of internal, switching, and leakage power. Vectorless power analysis is used, and assumes a signal toggle rate of 0.1, a clock toggle rate of two transitions per period, and a clock static probability of 0.5. Aprisa reports DRVs and shorts for Aprisa layouts. For ORFS layouts, an unnamed commercial P&R tool reports DRV and short counts after importing the routed DEF. Table 2 reports the post-route evaluation results for Aprisa and ORFS.

As is to be expected when compared to a commercial tool at a relatively aggressive clock period constraint, the ORFS flow averages 13.88% higher area, 26.68% higher wirelength, and 43.66% higher total power compared to the Aprisa flow. ORFS has worse setup WNS and setup TNS on 15 of the 18 benchmarks. ORFS has better setup timing for both ASAP7 Xiangshan and NG45 picorv32, so there is also an opportunity for Aprisa to improve further. ORFS achieves better hold timing closure for NG45. Some significant positive hold WNS values for ORFS suggest that ORFS may be over-fixing hold, or have a hold timing miscorrelation on the conservative side versus the commercial STA tool. While reported DRVs and shorts are reported by different tools, both Aprisa and ORFS have fairly low DRV and short counts on most designs, indicating good routing closure for both tool suites.¹

2.3 PDK Modifications and Reproducibility

In the benchmark enablement, Layers M1 through M3 of the ASAP7 technology LEF are modified to conform to the ASAP7 design rule manual and the original ASAP7 publication. These modifications correct the LEF rule definitions for minimum area, end-of-line spacing, width, grid alignment, and rectangle-only requirements. Rule definitions for layers M4 through M9 are unmodified.² The GitHub repository [23] contains the modified TechLEF files, the PDK-related flow enablements (e.g., Aprisa techfile), and the common inputs for every benchmark testcase. This enables reproduction of all flow outcomes (routed DEF and final gate-level netlist), which are also given in the repository.

3 Research Infrastructure and Enablements

Design automation researchers require strong and evolving foundations that enable discovery of relevant, reproducible advances in

¹The Aprisa flow setup targeted an outcome of no more than 100 post-route DRC violations – not a 0-DRC result.

²This fix has been submitted upstream as a pull request to the ORFS repository [24].

PDK	Design	Macros	Core Area (μm^2)	TCP (ps)
ASAP7	aes_cipher	0	2202	300
	ariane	136	74292	900
	blackparrot-2	24	101988	850
	jpeg_encoder	0	8727	300
	leon3_vta_bus	150	556611	900
	mempool_tile	20	89755	4000
	picorv32	0	2065	300
	tpu	0	7968	200
NG45	xiangshan	39	210359	1400
	aes_cipher	0	41038	500
	ariane	136	2200980	1300
	blackparrot-2	24	1973336	1300
	jpeg_encoder	0	125601	600
	leon3_vta_bus	150	17436679	1300
	mempool_tile	20	1536843	6800
	picorv32	0	36248	900
	tpu	0	115919	700
	xiangshan	39	4858982	2003

Table 1: Testcase characteristics including macro count, core area, and target clock period (TCP). These inputs are shared by Aprisa and ORFS.

the field. Developing such foundations in open source has been a long-standing focus of the DATC's efforts. In this section, we discuss two developments made in 2026: (i) a standard-cell generation methodology to improve the quality of open-source research PDKs, and (ii) the introduction of 3D OpenDB in OpenROAD to enable research on chiplet-based and 3D IC integration architectures.

3.1 Cell Generation for Academic PDKs

Open-source academic PDKs allow researchers to evaluate their designs and EDA research results on enablements that reflect real-world contexts – and to reproduce those results publicly. However, the cell libraries distributed with these PDKs support far fewer logic functions and drive strengths than industry libraries. Since synthesis and optimization can only search within the cell set that a library provides, this gap limits achievable design quality and relevance of experiments. Recent efforts address this limitation in the GT2N [29] and IHP SG13G2 [30] PDKs, each of which initially offers only a small set of standard cells.

Extending a given cell library can leverage recent solver-based cell layout synthesis tools, thus avoiding costly manual layout. Before that, a circuit topology (schematic, or CDL Spice netlist) must be available for any given new cell. Since the open-source ASAP7 cell library covers 212 cells, far more than what is available in other academic PDKs, we can extend a given target PDK by obtaining a cell topology from ASAP7, mapping it onto devices in the target PDK, and then generating the layout automatically using a solver setup matched to the target PDK.

(1) **GT2N.** We extend the GT2N cell library using SO3-Cell [31], which solves circuit topology, transistor placement, and in-cell routing together as a single mixed integer-linear program. We add 25 function/drive strength combinations, realized as 43 new cells across two GT2N nanosheet-width variants. 15 of these new cells realize 11 functions absent from the original library; the remainder add drive-strength variants to eight functions that are already present in the original library. We additionally illustrate how library augmentation methods can open the door to design-technology co-optimization or PPA exploration, by generating 86 new cells in a 7-track, 24nm-pitch geometry with a 43nm nanosheet, which the original 6-track library does not offer.

PDK	Design	Aprisa											ORFS										
		Cell		Setup			Hold			rWL	Power	DRV/Sh	Cell		Setup			Hold			rWL	Power	DRV/Sh
		Count	Area	WNS	TNS	FEP	WNS	TNS	FEP				Count	Area	WNS	TNS	FEP	WNS	TNS	FEP			
ASAP7	aes_cipher	19694	1904.6	-0.014	-0.73	163	0.034	0.00	0	87.7	11.11	0/0	15324	1505.5	-0.065	-7.95	160	0.031	0.00	0	66.9	10.44	0/0
	ariane	157811	37919.7	-0.050	-14.82	1878	0.028	0.00	0	1274.6	491.80	95/6	220983	46853.0	-0.294	-907.51	7102	0.029	0.00	0	1574.9	536.80	0/0
	blackparrot-2	24728	18246.5	-0.008	-0.04	19	0.009	0.00	0	465.4	64.98	11/0	162271	36524.2	-0.162	-9.62	190	0.015	0.00	0	1326.9	211.99	1/0
	jpeg_encoder	73706	7136.8	-0.072	-8.09	652	-0.002	-0.01	9	152.3	51.56	0/0	63955	7399.7	-0.067	-75.24	2784	0.013	0.00	0	158.5	96.01	0/0
	leon3_vta_bus	145997	296514.5	-0.090	-1.09	159	-0.009	-0.02	6	2195.8	446.43	12/0	296613	320051.6	-0.376	-599.58	4544	0.013	0.00	0	4013.0	901.74	0/0
	mempool_tile	195761	24779.5	-0.045	-18.83	1958	-0.018	-11.18	3882	1713.0	27.26	30/0	282264	36354.0	-0.063	-9.31	843	-0.036	-1.50	247	2100.4	42.80	0/2
	picorv32	14438	1515.5	-0.055	-2.17	242	-0.004	-0.01	2	64.8	11.98	15/1	15621	1822.7	-0.247	-202.00	1701	-0.041	-0.29	14	87.2	18.30	0/0
	tpu	70991	6424.4	-0.147	-95.80	1118	-0.015	-0.21	35	162.0	65.88	0/0	70762	7304.2	-0.291	-392.21	1985	0.003	0.00	0	204.6	131.42	0/0
	xiangshan	366393	77682.6	-0.041	-5.79	424	-0.001	0.00	2	3360.2	162.28	87/15	462912	95621.6	0.003	0.00	0	-0.039	-0.16	12	3833.9	259.18	9/1246
NG45	aes_cipher	22082	32549.9	-0.118	-13.76	309	0.107	0.00	0	329.1	42.50	6/6	19036	21861.7	-0.172	-22.12	162	0.092	0.00	0	244.3	31.33	1/0
	ariane	173164	1331378.4	-0.275	-123.78	3521	-0.025	-1.60	194	6775.2	785.41	29/1	162105	1360223.5	-1.181	-13037.45	21274	-0.004	0.00	1	7049.8	830.92	4/0
	blackparrot-2	127512	745470.3	-0.057	-46.23	2549	0.006	0.00	0	5451.7	478.18	14/0	206927	871508.3	-0.395	-1125.29	5120	0.018	0.00	0	6906.9	592.20	4/0
	jpeg_encoder	91338	111674.8	-0.022	-2.00	330	0.000	0.00	0	657.7	136.17	0/0	106258	111096.0	-0.117	-63.38	1164	0.016	0.00	0	860.1	146.78	1/0
	leon3_vta_bus	262493	9568619.6	-0.040	-7.20	818	-0.004	-0.01	2	18352.4	3743.19	12/0	286203	9697924.8	-1.687	-3274.36	8058	0.020	0.00	0	17139.9	4082.73	1/0
	mempool_tile	155271	469127.7	-0.104	-189.59	5011	-0.030	-6.95	606	6914.5	75.22	0/0	206383	560277.4	-0.248	-681.67	4957	-0.029	-0.85	72	7632.2	88.26	6/0
	picorv32	20481	29823.7	-0.213	-147.72	1288	-0.110	-5.65	77	268.9	28.34	1/1	26800	34231.3	-0.186	-58.33	815	0.045	0.00	0	425.9	34.49	2/0
	tpu	68336	80452.5	-0.032	-1.48	225	-0.015	-0.23	43	534.7	81.76	4/4	71787	84214.5	-0.132	-93.28	1085	0.010	0.00	0	567.5	104.86	1/0
	xiangshan	302499	1827605.1	-0.067	-92.09	4907	-0.050	-13.54	983	13950.9	598.62	38/0	352949	1933373.7	-0.757	-1083.29	11374	0.017	0.00	0	14899.8	669.34	9/0

Table 2: Post-route evaluation results for Aprisa and ORFS across nine testcases in ASAP7 and NG45. Cell Count denotes the number of standard-cell instances, and Cell Area denotes the total standard-cell and macro area in μm^2 . Both metrics exclude filler and well-tap cells. rWL and Power denote routed wirelength in mm and total power in mW, respectively. WNS and TNS are reported in ns, with WNS rounded to three decimal places and TNS to two decimal places. For both setup and hold, a negative WNS value indicates a timing violation. Setup and hold FEP denote the corresponding numbers of failing endpoints. The paired DRV/Sh entry reports routing-only design-rule violations and shorts.

(2) **IHP SG13G2.** To extend the IHP SG13G2 cell library, we combine CSyn-fp [33], which generates transistor placement from the cell netlist, with Au-MEDAL [32], which completes the layout via satisfiability modulo theories-based detailed-grid routing. We select this combination because Au-MEDAL handles the bidirectional routing that bulk-planar metal layers require. CSyn-fp and Au-MEDAL have been used to generate 28 cells that the original SG13G2 library does not provide. Eight new cells realize five missing functions (AOI31, OAI22, OAI31, AO22, OA21), and the remaining 20 add drive-strength variants to existing functions (e.g., INV/BUF to x32, NAND2/NOR2 to x4/x8, XNOR2 to x2/x4).

All generated cells for both PDKs pass DRC and LVS. (For IHP SG13G2, DRC includes both the main and maximal IHP SG13G2 rule sets; the unmet global density rules do not pertain to standalone cells.) For both PDKs, we convert the resulting GDS layouts to LEF using a conversion tool [34].

3.2 3D OpenDB

Chiplet and 3D IC design require a representation that preserves die hierarchy, heterogeneous technologies, bonding interfaces, and cross-die connectivity. We extend the OpenROAD database (ODB) with native chip-level objects for this purpose [44]. A dbChip represents a die, interposer, or hierarchical grouping, while a dbChipInst represents its placement and 3D orientation. Each die retains its own dbBlock and dbTech, allowing multiple process technologies to coexist in one assembly. An unfolded view maps the hierarchical assembly into global coordinates for physical analysis.

The same representation supports 3D visualization and structural checking, cross-die parasitic extraction using OpenRCX [45], and static timing analysis using OpenSTA [47]. OpenROAD also reads and writes 3Dblox descriptions [46], allowing these capabilities to operate on a common representation of the multi-die assembly. Current cross-die extraction combines conventional per-die OpenRCX

extraction with a separate model for the bonding interface. Ongoing work extends this interface toward geometry-based models for vertical-interconnect parasitics.

4 Advances in AI-Enabled EDA Research and Methodologies

AI-enabled EDA is one of the major research directions of OpenROAD-Research [38], the DATC-supported platform for developing and evaluating research extensions to OpenROAD and ORFS. This section describes representative works that have been developed as part of this effort, and whose implementations will be released through OpenROAD-Research to support reproducibility and follow-on research. More broadly, these works serve as exemplars of OpenROAD-Research's role in providing a common open-source environment for exploration of emerging AI-based methodologies across flow, algorithm, and source-code levels.

The following discussion reviews (i) A2-ORFO performs closed-loop optimization of an existing RTL-to-GDSII flow; (ii) AutoDRI [65] translates design rules into executable solver constraints for standard cell synthesis; (iii) ReviewDSE [60, 64] extends conventional *white-box design space exploration* (DSE) from exposed tool knobs to source-level optimization mechanisms; and (iv) GoalEvolve [61, 67] goes one step further by deriving mechanism-level code changes directly from residual QoR gaps. Then, as the agent action space moves deeper into the EDA stack, reliable design state and evaluation become increasingly important. We therefore conclude with discussion of (v) EDA-Schema-V2.1 and OpenROAD-ML-Injection, which provide versioned physical-design data and in-flow evaluation of learned models through their downstream impact on design QoR.

4.1 Agentic Stage-Aware Flow Optimization

A natural starting point for agentic EDA is the use of existing EDA tools. Traditionally, engineers inspect intermediate design states, adjust flow parameters, rerun the tools, and decide how subsequent stages should proceed based on the resulting QoR. This optimization

process is inherently stage-aware. A2-ORFO captures this stage-by-stage decision process in an agentic optimization loop. Rather than treating each RTL-to-GDSII run as a single black-box evaluation, it uses intermediate design states to decide which runs should continue, which should be terminated, and how promising states should be further explored. In each iteration, A2-ORFO launches $k=50$ parallel RTL-to-GDSII runs, prunes unpromising ones at two mid-stage checkpoints, and updates the parameter batch through large language model (LLM) reasoning coupled with Bayesian optimization (BO). All runs are evaluated under a fixed clock period CP_0 , minimizing $|TNS@CP_0|$ subject to a power non-degradation constraint $P \leq P_{base}$.

Multi-agent reasoning. A2-ORFO uses three collaborative agents to reason over historical QoR, feasibility, failures and lineage. *Agent1* (Surrogate Planner) employs a ReAct loop to configure the a Gaussian process regression surrogate according to observed prediction errors. *Agent2* (Hybrid Searcher) combines LLM-generated candidates, Bayesian sampling, local exploitation and global exploration, and uses the surrogate to select a diverse candidate set. *Agent3* (Textual Critic) reviews the execution outcome and rewrites the instructional prompts for the other two agents in the next iteration.

Doomed-runs prediction. A Transformer-based QoR predictor operates at the post-placement and post-CTS stages to estimate whether an ongoing run is likely to achieve favorable final timing and power. The predictor compares an ongoing partial trajectory with a complete reference trajectory using checkpoint QoR metrics, flow progress and clock information. It produces better/similar/worse predictions for final TNS and power, which are combined to rank the promise of ongoing runs.

Phased pruning and replication. A Go-With-The-Winners scheme selects candidates via Boltzmann-weighted ranking using the predictor's promising score; the selected candidates receive Boltzmann survival weights $w_i \propto \exp(-\beta \hat{r}_i)$ over the normalized rank $\hat{r}_i$, where β sets the exploitation strength. Then it propagates promising states through checkpoint-aware cloning that preserves upstream configurations while resampling downstream parameters. This design allows multiple descendants to inherit a promising physical state while exploring distinct downstream settings. All source-target propagation edges are lineage-tracked, and only full-trajectory survivors enter the optimizer's history to support reproducibility.

A2-ORFO is evaluated using three designs (AES, IBEX, JPEG) across ASAP7 and NG45. Each design is evaluated with 50 parallel ORFS instances per iteration over a total of 25 iterations. Experimental results show improved $|TNS@CP_0|$ and faster convergence with significantly fewer iterations than standard Bayesian methods.

4.2 Agentic Design Rule Integration

Stage-aware flow optimization assumes that the required EDA functionality is already available. A natural next step is to let agents assist with the implementation of explicitly specified design tasks. AutoDRI [65] explores this setting through design rule integration for constraint-based standard-cell synthesis. Constraint-based standard-cell synthesis encodes design rules as constructive CP-SAT constraints [66], thereby generating DRC-clean layouts. However, formulating natural-language design rules as solver constraints remains largely manual. AutoDRI uses LLMs to interpret natural-language design rules and formulate the corresponding CP-SAT constraints, reducing the manual effort required for rule integration.

To make this automation more robust, AutoDRI first reduces the semantic gap between high-level rule intent and low-level CP-SAT

variables. Its geometric semantic library maps primitive grid-edge variables to reusable semantic variables, including segment boundaries, metal-continuation and metal-empty intervals, and color-specific cut masks constructed for multicolor patterning. On top of this library, each design rule is normalized as a *conflict set*: a collection of forbidden local Boolean patterns, each representing one concrete violation scenario. This domain-specific intermediate representation gives the LLM a compact geometric vocabulary and a uniform output format, while decoupling rule understanding from solver-specific code generation. As shown in Figure 3, a Translator interprets a natural-language rule, enumerates its local violation scenarios, and expresses them as a conflict set; a Coder then compiles that conflict set into executable CP-SAT constraints within the synthesis engine. The Teacher-Translator loop compares generated semantics with demonstrated rules, diagnoses mismatches, and distills the resulting tips into reusable guidance. The generated engine is subsequently compiled and executed for cell synthesis, DRC and LVS. A Verifier aligns physical violations with the internal routing grid and separates semantic-coverage errors from coding errors, routing the former to a ConflictFixer and the latter back to the Coder.

The core idea of AutoDRI is to improve the reliability and consistency of LLM-agent-based rule integration through two complementary mechanisms: the semantic representation structures how agents reason about design rules, while execution feedback exposes missing violation cases and implementation errors. Together, they turn unconstrained one-shot generation into a closed-loop process whose outputs are semantically grounded and validated through synthesis and physical verification. AutoDRI is evaluated on 11 design rules, with three trials per rule. A trial was counted as successful only when the generated engine compiled and all 41 synthesized cells passed the target-rule check and LVS. GPT-5.4 and Gemini-3-pro succeeded in 32 and 33 of the 33 trials, respectively. The generated formulations remained comparable to manual hard-coding in runtime (an average ratio of $0.96\times$ on 14 complex cells), passed cell-level KLayout DRC and Cadence LVS, and eliminated cell-metal DRVs in three block-level studies with comparable PPA.

4.3 White-Box Design Space Exploration

Agentic design-rule integration still starts from human-specified functionality. White-box DSE takes a further step by allowing the agent to search how the EDA implementation itself should change. Conventional DSE treats the tool as a black box and explores only exposed parameters. Open-source EDA removes this boundary. In a codebase such as OpenROAD, source-level optimization mechanisms can be inspected, modified, and evaluated as part of the design space. ReviewDSE explores this setting by extending DSE from tool knobs to bounded source-level mechanisms.

Two-level exploration. ReviewDSE works at two levels. Level 1 runs calibration designs that expose different optimizer behaviors and freezes the warm-start material: it records which ideas improve results, which hurt, and which only add runtime, while source-start branches provide editable starting points. Level 2 explores the target design in a Teacher-Student loop. The Teacher assigns each Student a source start and a mechanism route; the Student edits a private workspace and submits a complete candidate; the Teacher reviews the full-flow outcome and promotes, repairs, or rejects it, keeping only reviewed results as mechanism-level evidence. The full framework is described in [64], with source code available in [60].

Protected evaluation. Uncontrolled edits can corrupt the evaluation path (stale binaries, tampered metric parsers) or bypass legality

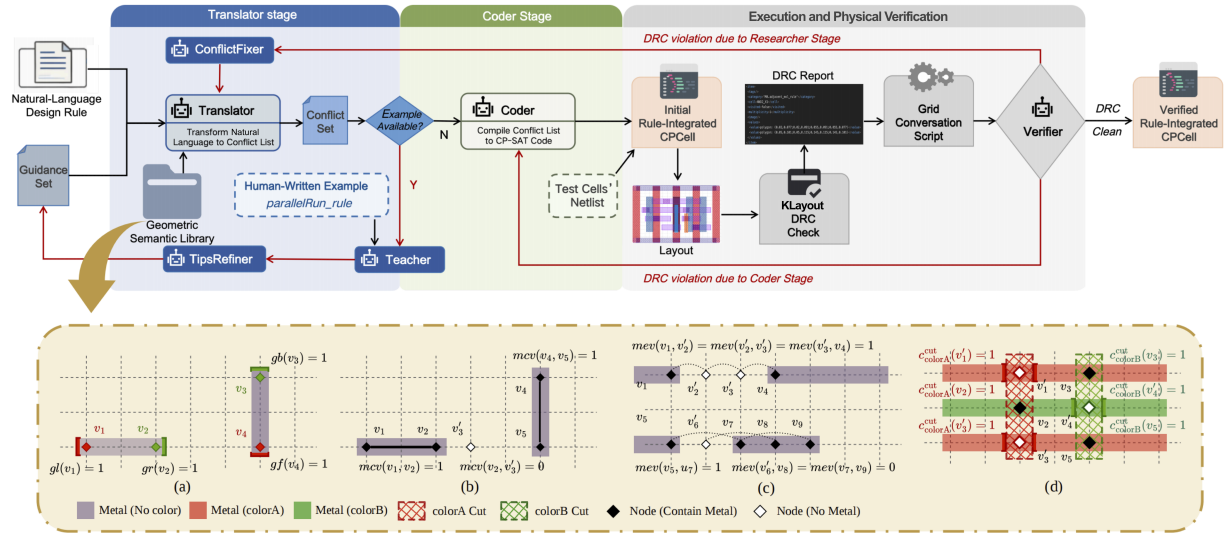


Fig. 3: Overview of AutoDRI. The upper part shows the feedback-driven multi-agent rule-integration flow, and the lower part shows the geometric semantic library that grounds LLM reasoning in the routed grid.

checks, and judging an edit only by its final result hides which mechanism caused a gain or where it broke later. ReviewDSE therefore fixes benchmark inputs, command sequence, legality checks, metric parsing, and runtime budget in a protected evaluator; candidates that modify or bypass this path are rejected.

Mechanism-level evidence. The core idea of ReviewDSE is to make source-level DSE reliable and cumulative. The protected evaluator acts as a fixed judge of quality, while mechanism-level evidence lets later searches build on reviewed results instead of starting over. ReviewDSE is evaluated on OpenROAD detailed placement, where legalization and detailed placement optimization interact and gains made in this stage can cause failures later in the flow. Across all nine target tasks [60], ReviewDSE consistently improves final placement quality over BO-based DSE under comparable runtime constraints.

4.4 Goal-Driven Algorithm Evolution

White-box DSE expands the search space from exposed tool parameters to source-level optimization mechanisms, but the search scope and mechanism direction are still largely predefined. GoalEvolve [67] moves one step further. It starts from a design-level QoR goal and determines what algorithmic mechanism should change next. Rather than relying on an expert to identify the bottleneck and formulate the next optimization idea, GoalEvolve uses the remaining QoR gap to drive mechanism discovery, source modification and full-flow evaluation. Figure 4 shows the overall evolution loop.

Goal-guided evolution planning. Based on the previous full-flow evaluation, GoalEvolve normalizes each metric’s residual gap to its target bound and identifies the dominant QoR bottleneck, which defines the planning objective for the next iteration. It traces the corresponding metric through stage-resolved checkpoints to locate where the deficit emerges or remains unrepaired and whether earlier gains persist. OpenROAD Cards then translate the diagnosed stage responsibility into a bounded source scope by linking each stage to its implementation path and required preservation constraints. Together, the bottleneck diagnosis and source mapping form the Teacher’s plan for the next revision.

Controlled evolution and attribution. Following the Teacher’s plan, four controlled Student branches realize alternative source-level hypotheses from a common parent. Two Explorers implement distinct new mechanisms, while an Enhancer strengthens a promising mechanism and an Integrator combines compatible validated mechanisms. Within each isolated branch, an LLM coding agent realizes the assigned mechanism in the bounded OpenROAD source, builds a private tool instance, and autonomously repairs implementation failures within a fixed budget. The resulting candidate is evaluated through the same checkpointed full-flow evaluation, allowing its outcome to be attributed to the assigned mechanism and retained as comparable evidence for subsequent evolution.

Mechanism memory and selection. After each round, the Evolutionary Program Database (EPD) records every Teacher idea with its implementation and evaluation evidence, assigning each mechanism a validated, promising, pending, or invalid status. For subsequent planning, the EPD filters records by compatibility with the current bottleneck and source state, then ranks them using accumulated evidence to guide subsequent evolution; negative evidence suppresses repeated ineffective proposals. The EPD also maintains a top- K portfolio across distinct mechanism families. The current champion normally seeds the next round, while plateau detection enables alternative parent selection without discarding the best program found.

GoalEvolve has been evaluated on post-placement timing and power optimization across eight ASAP7 designs, using commercial-tool results as reference QoR targets; code is available in [61]. GoalEvolve consistently improves the default OpenROAD flow and reduces the gap to the reference targets. These results show that design-level QoR goals can directly guide the discovery and evolution of source-level optimization mechanisms.

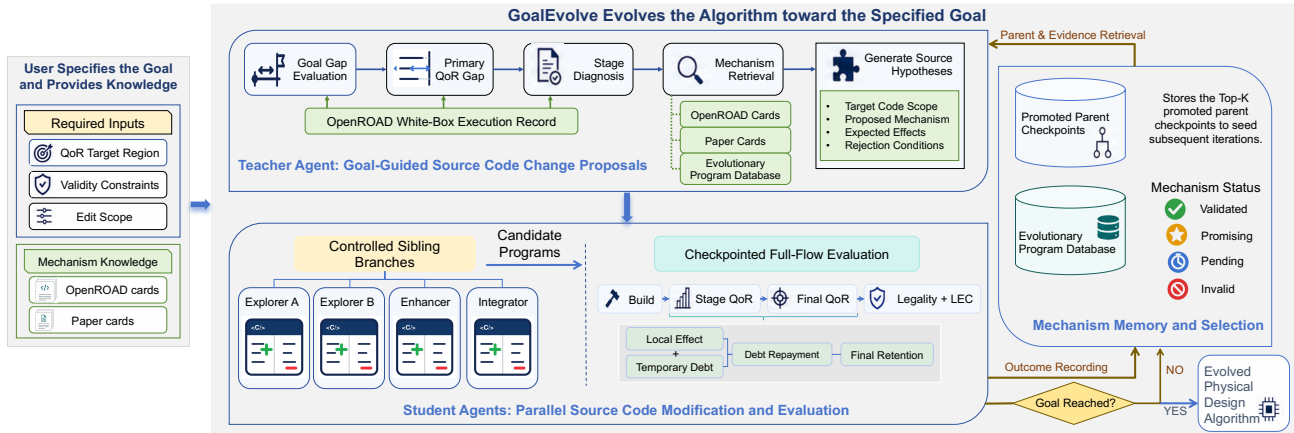


Fig. 4: Overview of GoalEvolve. The Teacher converts QoR gaps and retrieved knowledge into targeted source hypotheses, while parallel Students evaluate candidates through checkpointed full-flow runs. Reviewed outcomes update mechanism evidence and parent selection for subsequent iterations.

4.5 Versioned Physical-Design Dataset and Closed-Loop ML Evaluation

In RDF-2024, the DATC highlighted common data formats, datasets, and tool interfaces as important foundations for open and reproducible ML EDA research. RDF-2026 has incorporated EDA-Schema-V2 [49], which extends the graph-based representation and open dataset of EDA-Schema [48] with stage-resolved physical-design data and versioned dataset releases. V2 contains 7,776 implementations of eighteen IWLS'05 circuits [50] across NG45, SkyWater130 [51], IHP SG13G2 [30], and ASAP7, and defines twelve prediction tasks with stage-specific OpenROAD estimates as reference baselines. The schema, datasets, release metadata, and manifests are maintained through the EDA-Schema repository and open-dataset page [53, 54].

V2 was generated using ORFS commit a50fb65 [52]. EDA-Schema-V2.1 retains the same data model and prediction tasks while regenerating a representative subset using ORFS commit c2c8aee [55]. Designs are retuned near timing closure and regenerated in the Barely Fail (BF), $-10\% \leq \text{SCPR} < 0\%$, and Barely Pass (BP), $0\% \leq \text{SCPR} \leq 10\%$, regions, where SCPR (Slack-to-Clock-Period Ratio) equals $\text{WNS}/\text{Clock Period} \times 100$. Comparing V2 and V2.1 captures shifts in timing closure, flow completion, PPA, wirelength, routability, and stage-specific OpenROAD estimates as OpenROAD and ORFS evolve. Future releases will add macro-containing designs, additional benchmark circuits, and periodic dataset regeneration with newer tool versions.

Using the V2.1 dataset, an evaluation harness has been developed to evaluate trained models within the physical design flow [56]. OpenROAD-ML-Injection is a fork of ORFS-Research that exposes model injection points, allowing trained models to replace or augment selected OpenROAD estimates during execution. The community can replace the provided models with their own and compare matched baseline and model-guided ORFS runs. This enables models to be evaluated not only by standalone accuracy, but also by their impact on downstream optimization metrics, including PPA, WNS, TNS, wirelength, routability, design-rule violations, runtime, and flow completion. Thus, optimization outcomes directly measure whether a prediction is useful in the EDA flow, connecting versioned ML datasets to closed-loop evaluation of downstream QoR.

5 Community and Ecosystem

5.1 Workforce and Global Initiatives

Accessible courseware and training materials provide foundations for workforce development and broadened participation in EDA research. The growing scope and adoption of open-source EDA tools creates new opportunities for hands-on education using complete and reproducible design flows. To support development of and cross-pollination within this ecosystem, the Open-Source EDA Education Awesome List [57] catalogs courses and training programs built on open-source EDA. By consolidating freely available educational resources, the list helps to lower the barrier for students and researchers to learn modern EDA tools and methodologies.

Open-source EDA is also receiving increasing attention across a range of international initiatives. In the European Union, efforts including Chip Design Germany/DE:Sign [71] EuroCDP [69], and ODE4EC [70] emphasize the development and adoption of open-source EDA tooling and design enablements [58]. Complementary efforts address interoperability and orchestration across EDA tools and workflows. For example, EDA² [59] develops Python-based object models for interacting with RTL source and reports, and introduces abstractions for interacting with EDA tools and workflows. These education and interoperability efforts complement DATC's research infrastructure by making open EDA environments easier to learn, integrate and reuse across the community.

5.2 Contests

The support of leading academic contests is part of DATC's broadened scope of research enablement. Here, we highlight two 2026 contests on post-detailed placement timing optimization with close ties to RDF infrastructure: the ISPD 2026 Contest [17] and the MLCAD 2026 Contest [21]. Both contests use ASAP7 designs and provide environments for developing and evaluating timing optimization methods.

(1) **ISPD 2026 Contest.** The ISPD 2026 Contest focuses on gate sizing, V_t swapping, and buffer or inverter insertion under realistic physical-design constraints. Its benchmark suite [35] contains eight public and four hidden designs ranging from 15K to 1.4M instances. The benchmarks include macro-containing designs, hard

and soft placement blockages, and power-delivery-network constraints, exposing optimization methods to congestion and legalization constraints representative of practical implementation flows. The contest will be continued at ISPD 2027 with an expanded scope and updated benchmarks.

(2) **MLCAD 2026 Contest.** The MLCAD 2026 Contest considers gate sizing, V_t swapping, buffer or inverter insertion, pin swapping, and logical restructuring. It emphasizes *algorithm discovery* and *design-adaptive optimization* [68], allowing optimization strategies to be specialized to individual designs. The benchmarks range from approximately 20K to 310K instances and are organized into alpha and incrementally released beta sets.

Both contests address the domain of timing- and electrical rules-driven netlist optimizations, i.e., the functionality seen in OpenROAD's Resizer [37]. OpenROAD is used for evaluation, including global routing, parasitic extraction, and OpenSTA-based timing and power analysis. The contests also use a common Purdue Anvil environment [36] for development and official evaluation, providing a consistent computational setup.

6 RDF Roadmap Update

A year ago, RDF-2025 [16] outlined priorities spanning continuous integration, calibration and baselines, research enablement, reproducibility, and broader community engagement. Several of these directions have been advanced through the 2026 activities described in this paper. Based on this progress and the new capabilities and research directions that have emerged, we identify the following near-term focus areas and longer-term community challenges.

6.1 Near-Term Focus Areas

RDF and Continuous Integration. Near-term efforts will strengthen CI and QoR regression testing across representative testcases and PDKs. RDF will also broaden support for incorporating and preserving new research tools, including binary-only tools, and will begin incorporating chiplet and 3D IC capabilities.

Research Infrastructure and Enablement. OpenROAD-Research [38] and ORFS-Research [39] will expand community participation through clearer and more consistent contribution guidelines. Technology enablements will continue to mature through further development of 3D OpenDB and updates to open PDKs and their supporting design collateral. Notably, in the near term we will further extend OpenROAD-Research toward an end-to-end RTL-to-GDSII implementation flow [72] for face-to-face 3D integration in the LogicFold paradigm, with native 3D optimization incorporated throughout the flow. The flow will integrate 3D-aware synthesis, tier partitioning and placement, hybrid bonding terminal optimization, cross-tier timing optimization, clock tree synthesis, and native 3D routing within a unified and reproducible framework.

Benchmarking and Baselines. Building on the RTL-to-GDS benchmarking methodology in Section 2, near-term efforts will establish a maintained and repeatable benchmarking suite. Common testcase definitions and evaluation methodologies will be preserved, and results will be periodically refreshed as tools and flows evolve. The scope will also expand to additional designs, PDKs, and tools – with an eye to increased problem scale and design complexity – while maintaining reproducible reference and best-known results.

Evaluation Infrastructure for AI-Enabled EDA. Near-term efforts will develop common evaluation harnesses for AI-enabled and agentic EDA (see, e.g., [18]). These environments will fix benchmark inputs, objectives, validity checks, metrics, and downstream QoR

evaluation while clearly defining the portions of a flow or tool that an autonomous method is permitted to modify.

6.2 Longer-Term Community Challenges

Complete design enablements at advanced nodes. The community lacks a coordinated effort toward a single, complete, and commonly adopted open-source PDK with design enablement that is representative of an advanced node. Necessary enablement includes standard cells, memories, I/Os, and supporting technology collaterals consistent with industry-relevant fundamental EDA research.

Representative public designs and testcases. Larger and more realistic public designs and testcases are needed to better capture the constraints, optimization challenges, and behaviors encountered in industrial design.

End-to-end chiplet and 3D design flows and testcases. The community still needs broadly accessible flows, benchmarks, and testcases that exercise heterogeneous multi-die design, bonding interfaces, cross-die multiphysics analyses, and physical optimization.

Reproducibility and artifact evaluation for agentic EDA. Artifact evaluation approaches highlighted in [15] become more challenging for agentic EDA and agentic chip design. Results can depend on model versions, agent trajectories, tool access, and nondeterministic behaviors – even as reproducing an experiment can require substantial compute, token usage, runtime, and monetary cost. New approaches are therefore needed to make agentic artifacts reproducible, comparable, and practical to evaluate.

7 Conclusion

The 2026 DATC activities reflect a continued broadening of research enablement across design flows, benchmarking, technology infrastructure, and AI-enabled EDA. RDF continues to expand with new design-flow capabilities, while commercial benchmarking now extends from netlist-to-GDS P&R to RTL-to-GDS SP&R evaluation using common inputs and signoff methodologies. Research infrastructure has advanced through improved open PDK collaterals, 3D IC support, and continued development of OpenROAD-Research and ORFS-Research. AI-enabled EDA is progressing toward agentic flow optimization, source-level exploration, and autonomous evolution of EDA algorithms. Important challenges remain in complete advanced-node design enablements, representative public testcases, end-to-end chiplet and 3D design flows, and reproducible and practical evaluation of agentic EDA artifacts. These directions will continue to shape DATC efforts over the coming year, toward lowering critical barriers to EDA research.

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